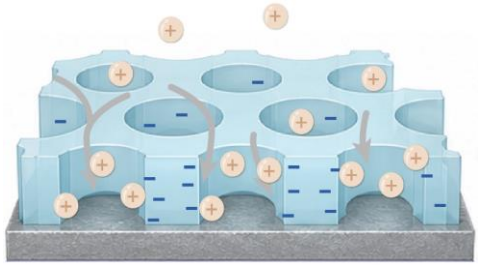
Research Overview-Jinhao Li

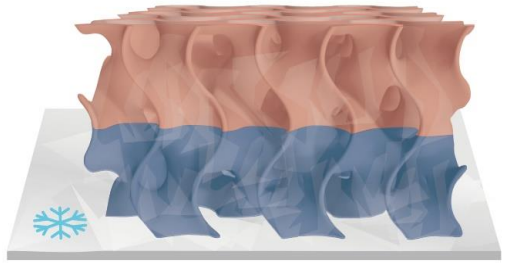
Three-dimensional sensing iontronics

Disordered hysteretic ion transport

Architected elastic interfaces



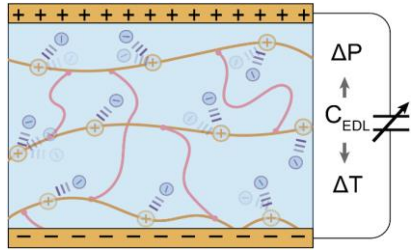
Charge redistribution



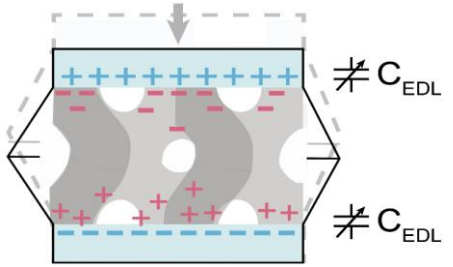
Iontronic metastructure

Nonlinear coupled iontronic behaviors

Multisensory decoupled devices



Multimodal decoupling

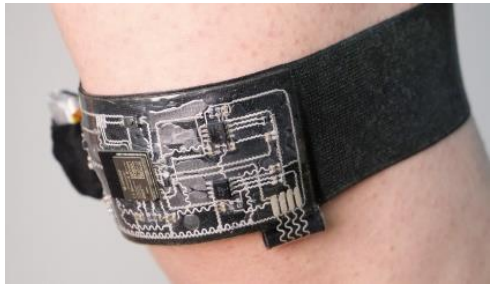


Nonlinearity decoding

Integrated applications



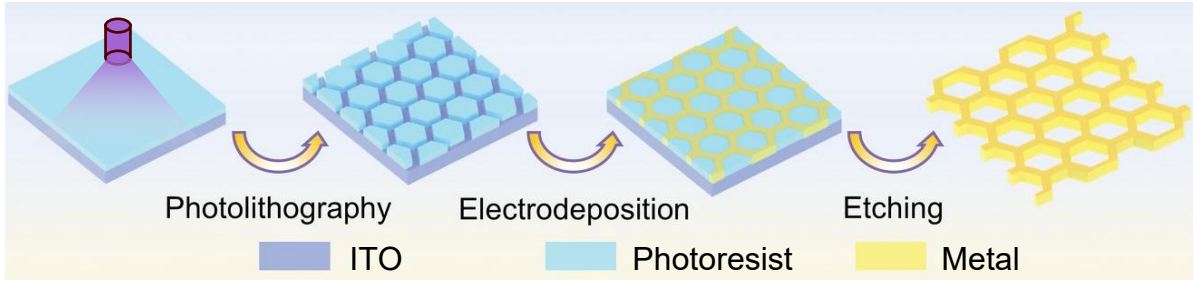
Sensorized prostheses

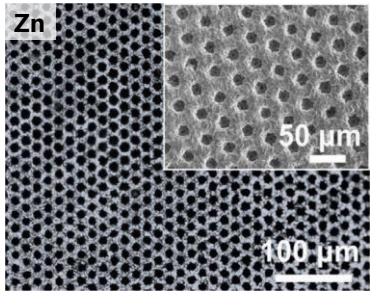


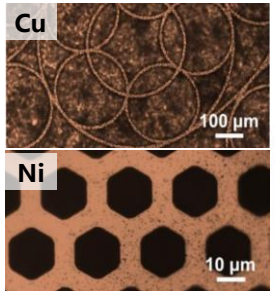
Wearable electronics

Architected elastic interfaces

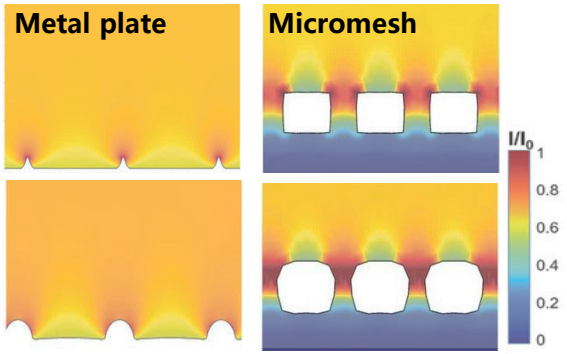
Micromesh-guided charge redistribution



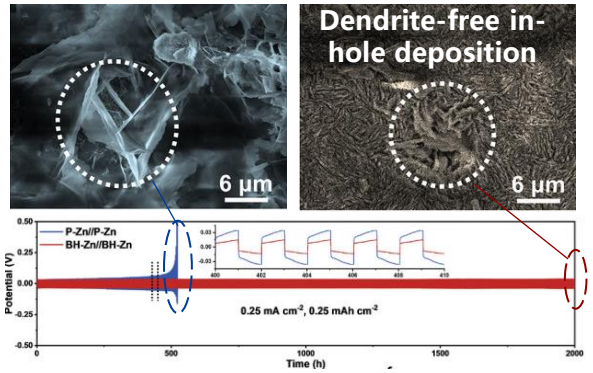






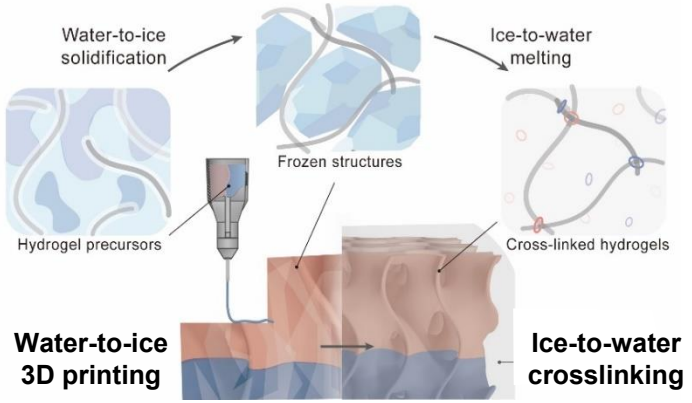


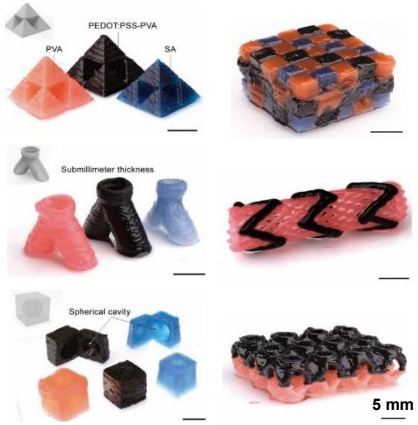
Geometry-induced charge redistribution

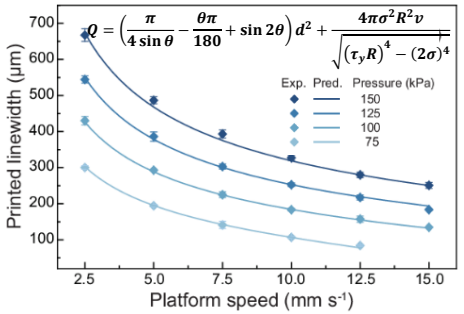


Interfacial stability increased by 280%

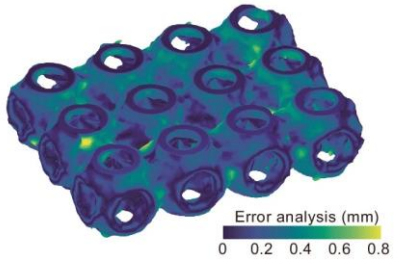
Cryogenic printing iontronic metastructure

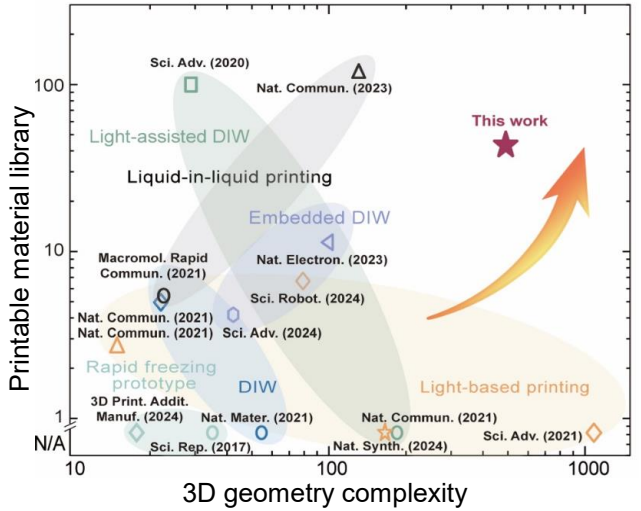






High-resolution fabrication



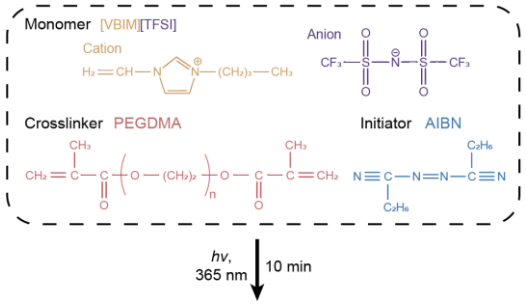


Extreme high-aspect ratio (2 cm/42 μm=476)

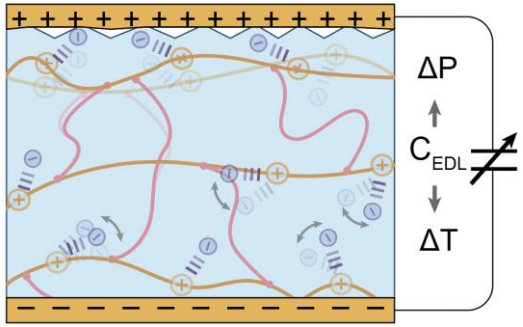
Mechanical tunability (3.7-153.6 kPa)

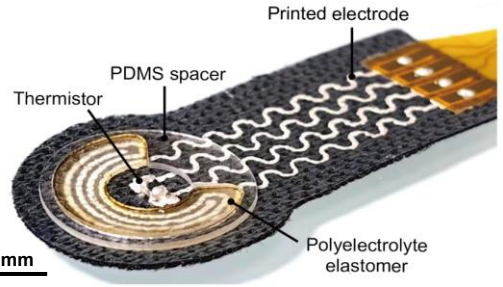
Multisensory decoupled devices

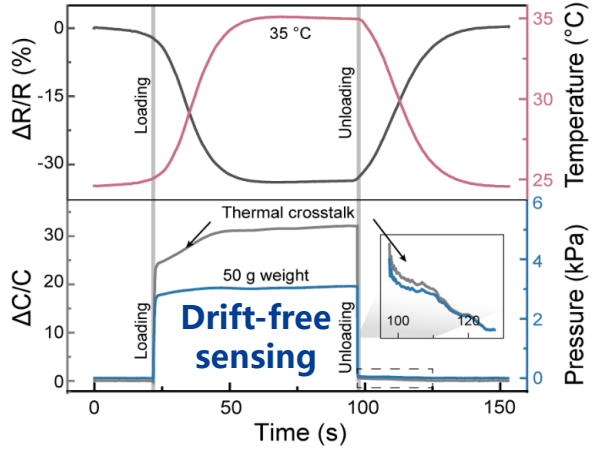
Pressure-temperature decoupling

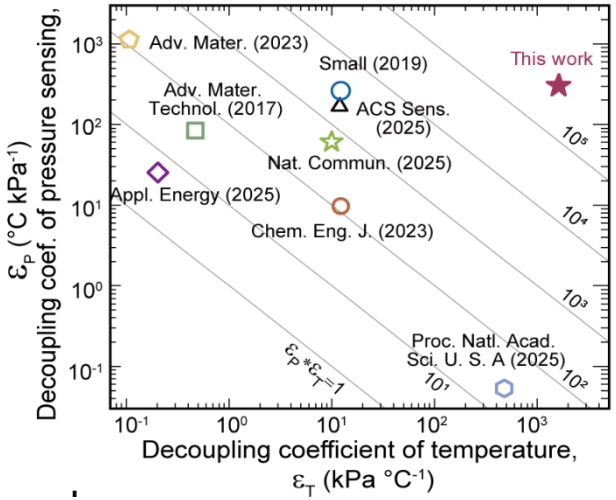


Covalent polyelectrolyte elastomer decouples thermal ionic polarization



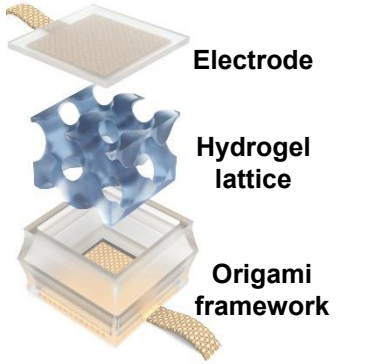


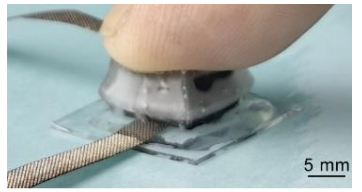


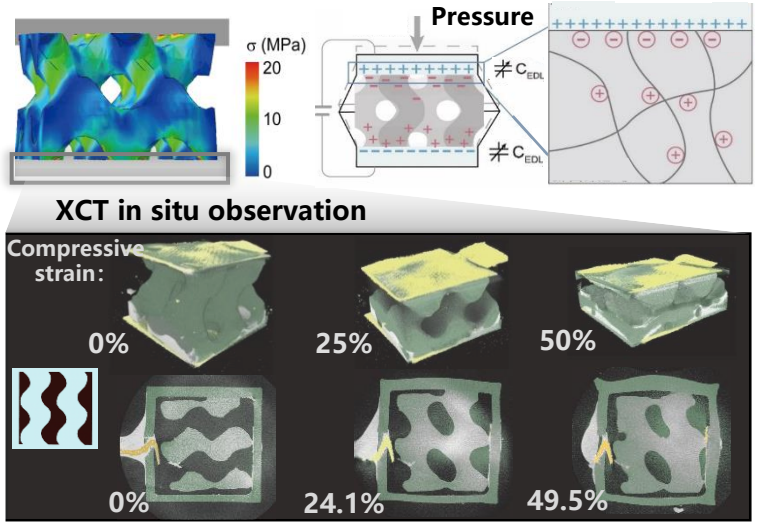


Highly-decoupled multisensory

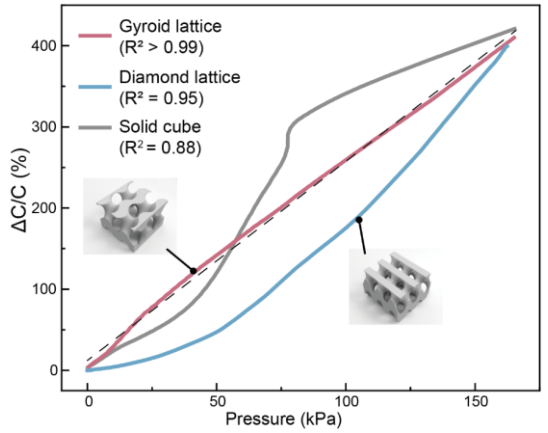
Iontronic-mechanical linearizing

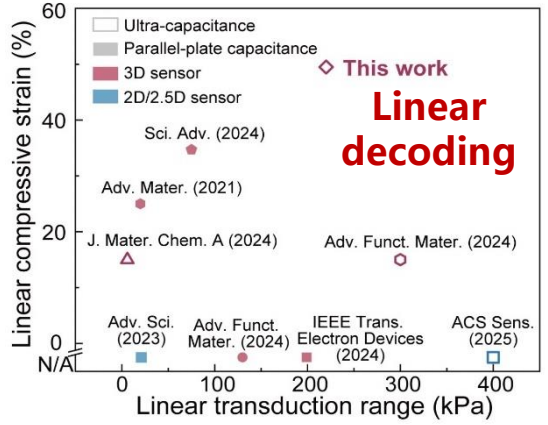






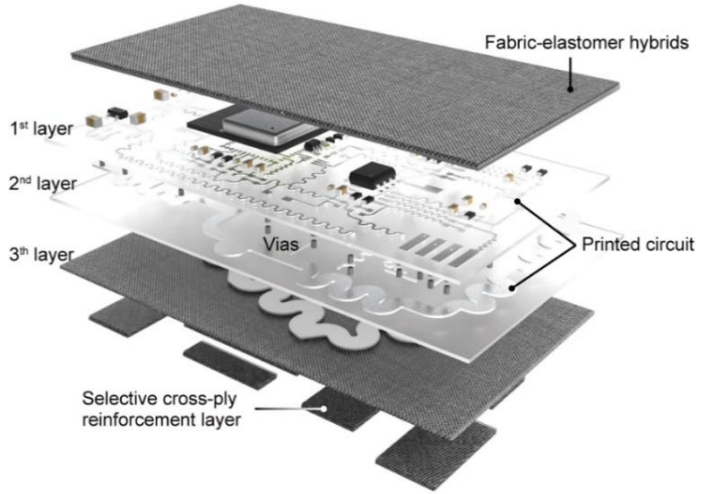
Strain ε ∝ Contact $\Delta S/S$ ∝ Responses $\Delta C/C$

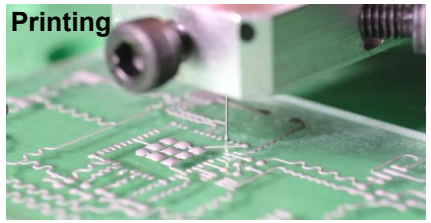


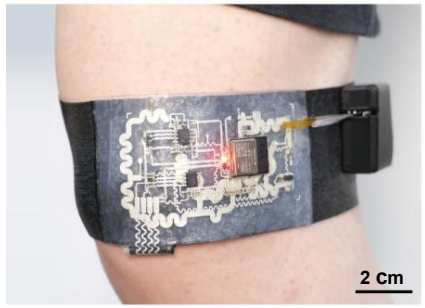


Integrated applications-Wearable electronics

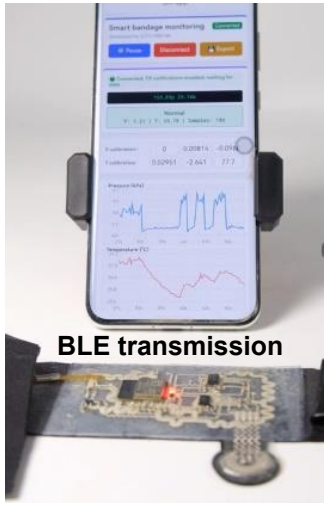
Multisensory electronic bandage

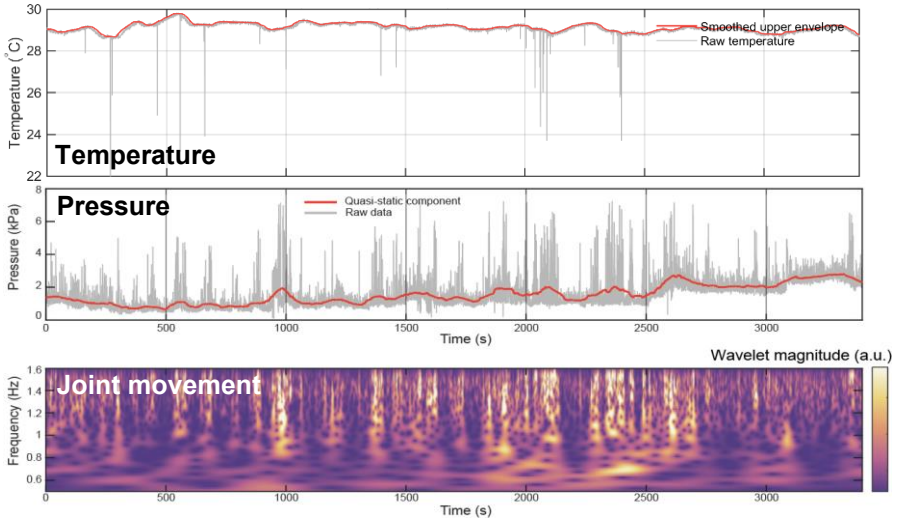




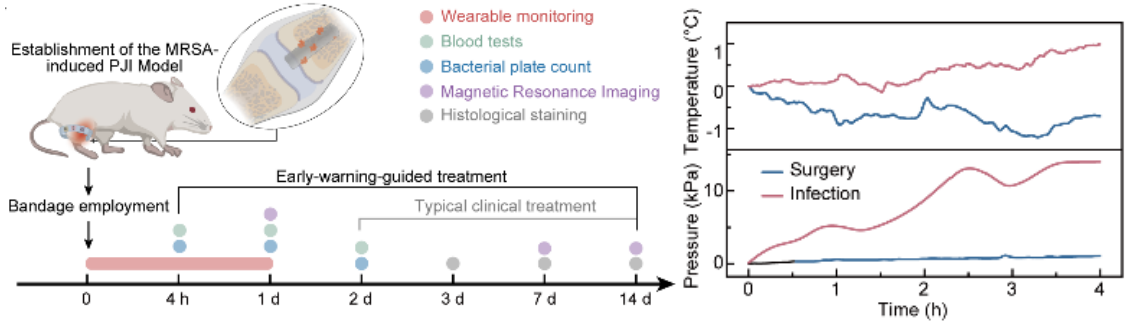


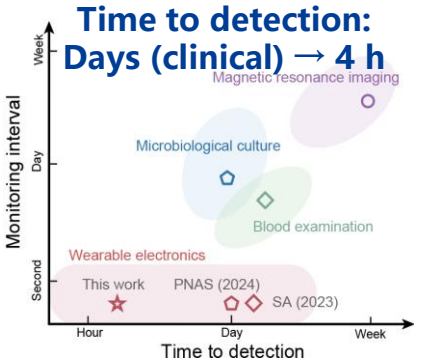
Multilayer stretchable processing circuit

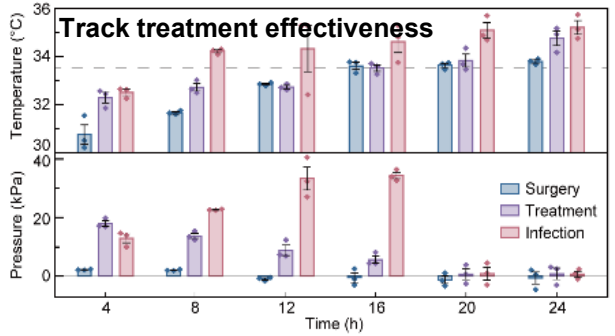


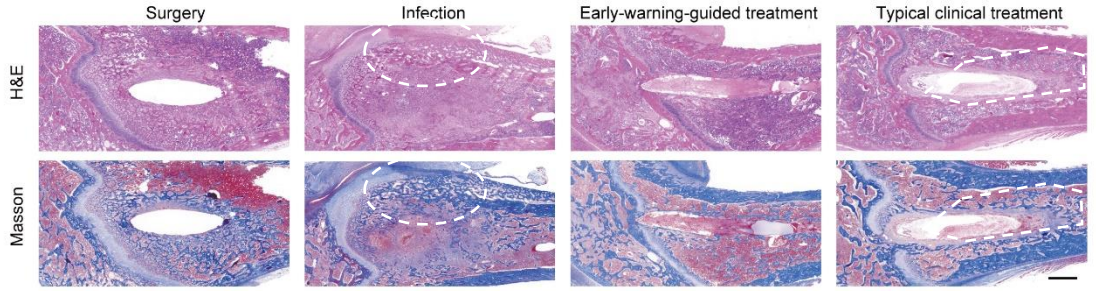


Early warning of periprosthetic joint infection





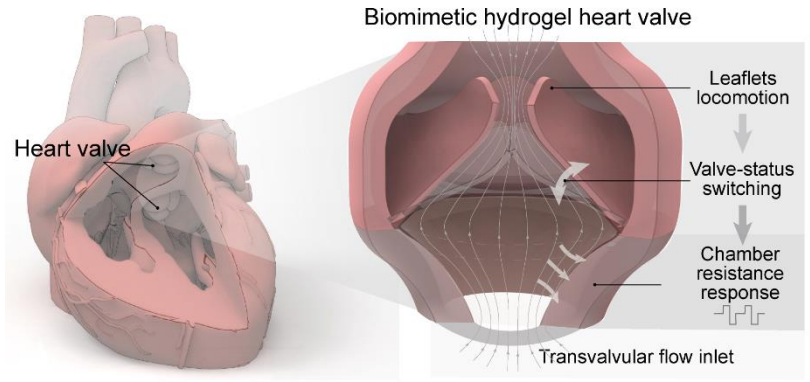


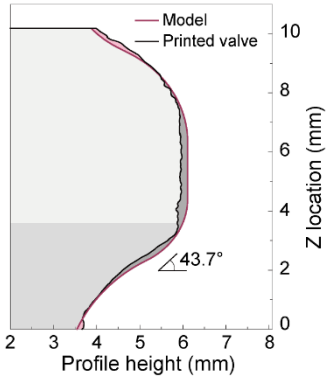


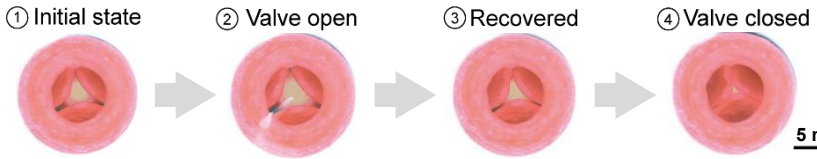
Early-warning-guided intervention reduces tissue damage

Integrated applications-Sensorized prostheses

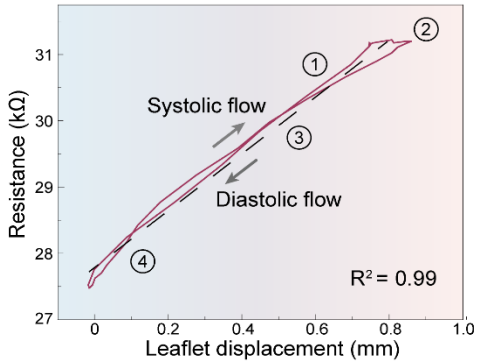
Self-sensing hydrogel-based aortic valve

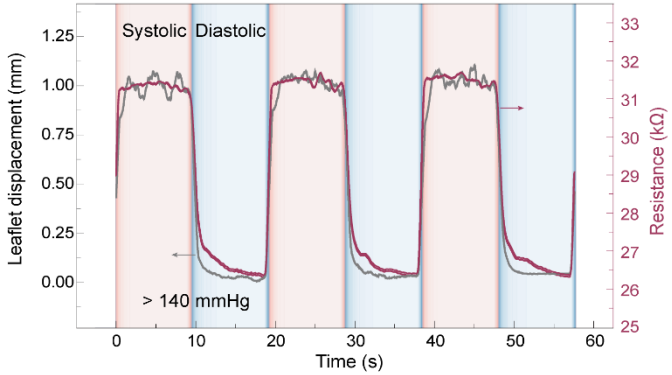






Flow-driven tricuspid valve to open/close



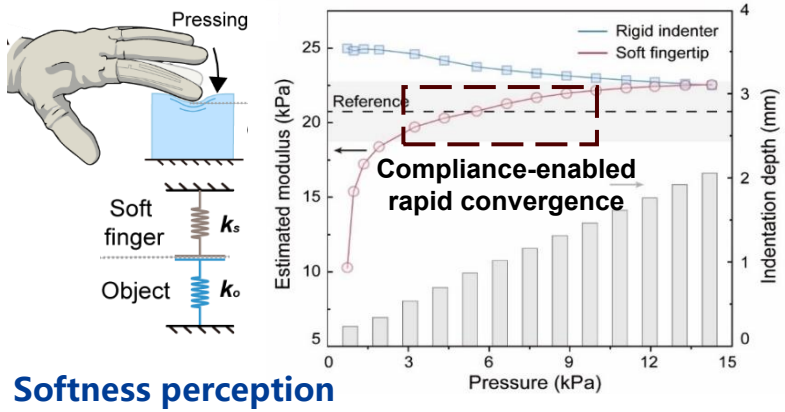


- High linearity (R^2=0.99)
- Low hysteresis (~8.3%)

Proprioception of soft bioprosthesis

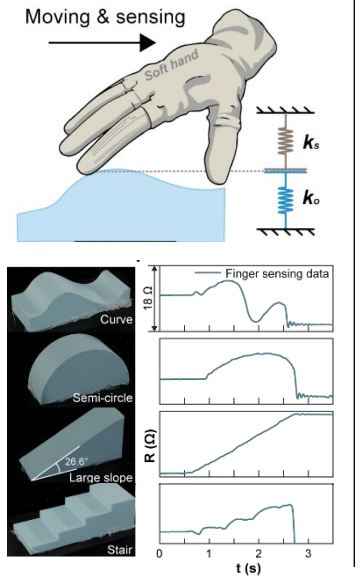
Somatosensory soft neuroprosthetic hand



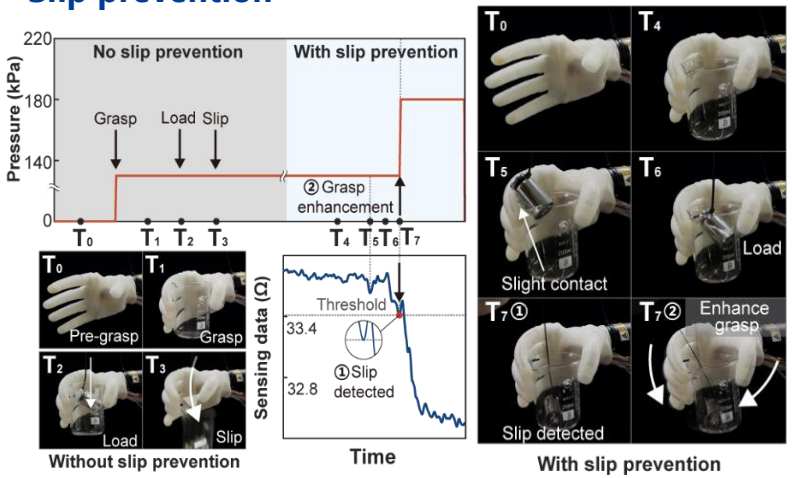


Softness perception

Shape detection



Slip prevention



Closed-loop control based on tactile and proprioceptive feedback